```
#include <LiquidCrystal.h>
#include <SoftwareSerial.h>
#include <OneWire.h>
#include <DallasTemperature.h>
#define USE_ARDUINO_INTERRUPTS true
#include <PulseSensorPlayground.h>
SoftwareSerial esp(10, 11);
LiquidCrystal lcd(7, 6, 5, 4, 3, 2);
#define ONE_WIRE_BUS 9
#define TEMPERATURE_PRECISION 12
OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);
DeviceAddress tempDeviceAddress;
int numberOfDevices, temp, buzzer = 8;
const int PulseWire = A0;
int myBPM, Threshold = 550;
PulseSensorPlayground pulseSensor;
unsigned long previousMillis = 0;
const long interval = 5000;
void setup()
{
  lcd.begin(16, 2);
  Serial.begin(9600);
  esp.begin(115200);
  sensors.begin();
  numberOfDevices = sensors.getDeviceCount();
  pulseSensor.analogInput(PulseWire);
  pulseSensor.setThreshold(Threshold);
  pulseSensor.begin();
```

```
  pinMode(buzzer, OUTPUT);
  digitalWrite(buzzer, HIGH);
  lcd.setCursor(0, 0);
  lcd.print("  IoT Patient");
  lcd.setCursor(0, 1);
  lcd.print(" Monitor System");
  delay(1500);
  digitalWrite(buzzer, LOW);
  lcd.clear();
}

void loop()
{
  myBPM = pulseSensor.getBeatsPerMinute();
  if (pulseSensor.sawStartOfBeat())
  {
    beep();
    lcd.setCursor(0, 1);
    lcd.print("HEART:");
    lcd.print(myBPM);
    lcd.setCursor(9, 1);
    lcd.print(" BPM");
    delay(20);
  }
  sensors.requestTemperatures();
  for (int i = 0; i < numberOfDevices; i++)
  {
    if (sensors.getAddress(tempDeviceAddress, i))
    {
```

```
      temp = printTemperature(tempDeviceAddress);
      lcd.setCursor(0, 0);
      lcd.print("BODY:");
      lcd.print(temp);
      lcd.print(" *C");
    }
  }
  upload();
}

int printTemperature(DeviceAddress deviceAddress)
{
  int tempC = sensors.getTempC(deviceAddress);
  return tempC;
}

void beep()
{
  digitalWrite(buzzer, HIGH);
  delay(150);
  digitalWrite(buzzer, LOW);
}

void upload()
{
  unsigned long currentMillis = millis();
  if (currentMillis - previousMillis >= interval)
  {
    previousMillis = currentMillis;
```

```
    esp.print('*');
    esp.print(myBPM);
    esp.print(temp);
    esp.println('#');
  }
}
```